

21EIC552J – Industrial Process Automation

Question Bank

Module-1 - Automation System Structure

1. List the evolution of automation technology over the past decades
2. Define Physical process and name its categories
3. What is a process signal and list the types of physical signals.
4. List the benefits of Automation
5. Draw the basic structure of automation system
6. Draw the Typical automation cycle.
7. What are the tasks performed by Control Subsystem
8. Describe Process Management system through the schematic and operation of the unattended/ attended/ automated water heater system.
9. Explain briefly about the automation steps in water-heating process
10. Discuss in detail about Input, output and human interface Instrumentation Subsystem with an example
11. Draw the structure of analog instrumentation device and explain its measurement and control mechanism.
12. Write the basic function of the human interface system
13. What is a control subsystem and its functions?
14. What is called an automation function
15. Define delay time in a process control
16. List the basic strategies for implementing automation schemes
17. What is meant by steady-state error
18. Discuss about various types of closed loop control schemes for continuous process automation
19. Illustrate with a neat sketch on Feed-Forward and Cascade control schemes.
20. Illustrate the interfacing of the control subsystem with the instrumentation subsystem with an example
21. Illustrate with the schematic an application PID control in an automobile
22. Describe about the remote management of communicable data acquisition and control unit with an example
23. Discuss about various types of closed loop control schemes for continuous process automation

Module-2 - PLC Programming and Troubleshooting

1. Express each of the following equations as a ladder logic program:

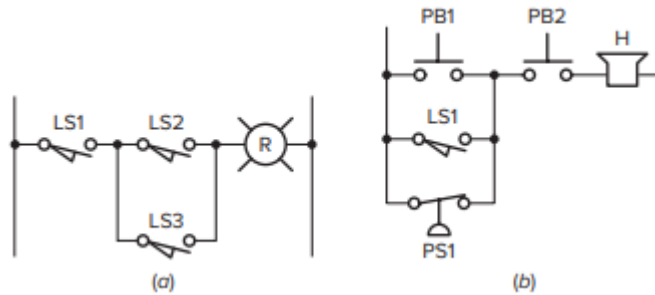
a. $Y = (A + B)CD$

b. $Y = A\bar{B}C + \bar{D} + E$

c. $Y = [(\bar{A} + \bar{B})C] + DE$

d. $Y = (\bar{A}BC) + (D\bar{E}F)$

2. Write the ladder logic program, draw the logic gate circuit, and state the Boolean equation for the two relay ladder diagrams in Figure



3. It is required to have a pilot light come on when all of the following circuit requirements are met:

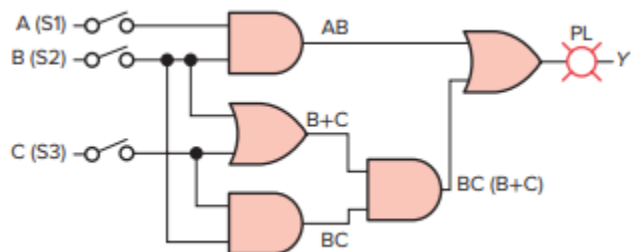
- All four circuit pressure switches must be closed.
- At least two out of three circuit limit switches must be closed.
- The reset switch must not be closed.

Using AND, OR, and NOT gates, design a logic circuit that will solve this hypothetical problem and also draw the ladder program.

4. For the logic gate circuit shown in Figure,

- a. Determine the Boolean equation.

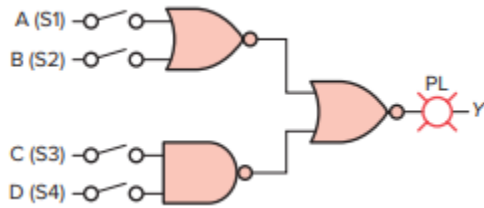
- b. Draw an equivalent ladder logic program for the gate circuit.



5. For the logic gate circuit shown in Figure,

- a. Determine the Boolean equation.

b. Draw an equivalent ladder logic program for the gate circuit.



6. Write the Boolean expression and draw the gate logic diagram and typical PLC ladder logic diagram for a control system wherein a fan is to run only when all of the following conditions are met:

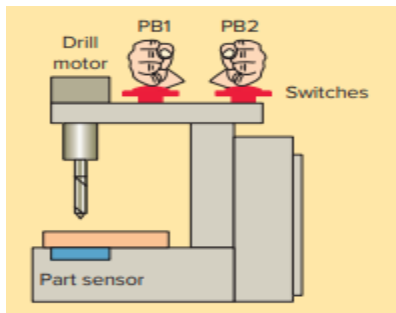
- Input A is OFF
- Input B is ON or input C is ON, or both B and C are ON
- Inputs D and E are both ON
- One or more of inputs F, G, or H are ON

7. Outline the sequence of events involved in a PLC scan cycle.

8. List the five standard PLC languages as defined by the International Standard for Programmable Controllers, and give a brief description of each.

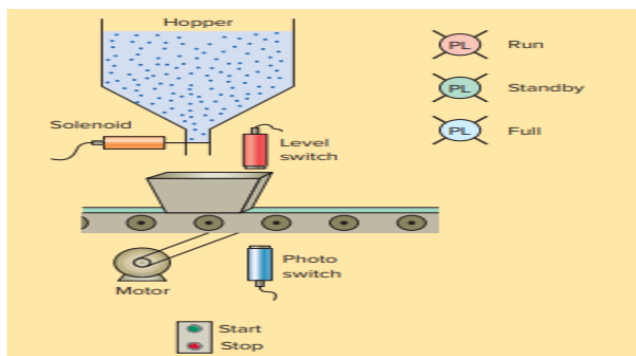
9. A normally closed pushbutton is connected to a PLC discrete input. Does this mean it must be represented by a normally closed contact in the ladder logic program? Explain why or why not.

10. Figure shows the sketch of a drilling process that requires the drill press to turn on only if there is a part present and the operator has one hand on each of the start switches. This precaution will ensure that the operator's hands are not in the way of the drill. The sequence of operation requires that switches 1 and 2 and the part sensor all be activated to make the drill motor operate. Develop the ladder logic program required for the process.



11. Figure shows the sketch of a continuous filling operation. This process requires that boxes moving on a conveyor be automatically positioned and filled. The sequence of operation for the continuous filling operation is as follows:

- Start the conveyor when the start button is momentarily pressed.
- Stop the conveyor when the stop button is momentarily pressed.
- Energize the run status light when the process is operating.
- Energize the standby status light when the process is stopped.
- Stop the conveyor when the right edge of the box is first sensed by the photo sensor.
- With the box in position and the conveyor stopped, open the solenoid valve and allow the box to fill. Filling should stop when the level sensor goes true.
- Energize the full light when the box is full. The full light should remain energized until the box is moved clear of the photo sensor.



A pump is to be used to fill two storage tanks. The pump is manually started by the operator from a start/stop station. When the first tank is full, the control logic must be able to automatically stop flow to the first tank and direct flow to the second tank through the use of sensors and electric solenoid valves. When the second tank is full, the pump must shut down automatically. Indicator lamps are to be included to signal when each tank is full. a. Draw a sketch of the process. b. Prepare a typical PLC program for this control process.

12. What is a seal-in circuit?

13. A temperature control system consists of four thermostats controlling three heating units. The thermostat contacts are set to close at 50°, 60°, 70°, and 80°F, respectively. The PLC ladder logic program is to be designed so that at a temperature below 50°F, three heaters are to be ON. From 50° to 60°F, two heaters are to be ON. For 60° to 70°F, one heater is to be ON. Above 80°F, there is a safety shutoff for all three heaters in case one stays on because of a malfunction. A master switch is to be used to turn the system ON and OFF. Prepare a typical PLC program for this control process.

14. 15. Design a PLC program and prepare a typical I/O connection diagram and ladder logic program for the following motor control specifications:

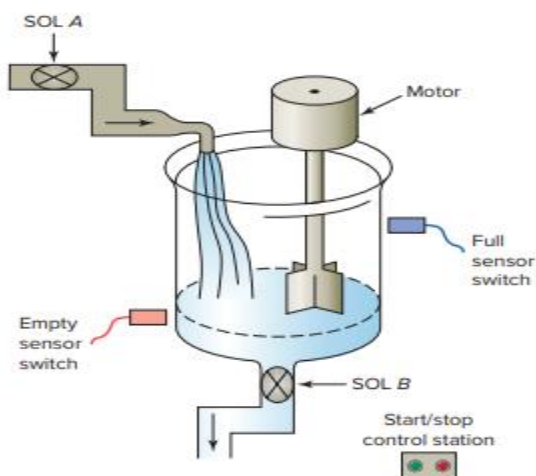
- A motor must be started and stopped from any one of three start/stop pushbutton stations.
- Each start/stop station contains one NO start pushbutton and one NC stop pushbutton.
- Motor OL contacts are to be hardwired.

16. a. Explain the difference between the operation of a nonretentive timer and that of a retentive timer. b. Explain how the accumulated count of programmed retentive and nonretentive timers is reset to zero.

17. Write a ladder logic program that will turn on a light, PL, 15 s after switch S1 has been turned on.

18. Write a program to implement the process illustrated in Figure. The sequence of operation is to be as follows:

- Normally open start and normally closed stop pushbuttons are used to start and stop the process.
- When the start button is pressed, solenoid A energizes to start filling the tank.
- As the tank fills, the empty level sensor switch closes.
- When the tank is full, the full level sensor switch closes
- Solenoid A is de-energized.
- The agitate motor starts automatically and runs for 3 min to mix the liquid.
- When the agitate motor stops, solenoid B is energized to empty the tank.
- When the tank is completely empty, the empty sensor switch opens to de-energize solenoid B.
- The start button is pressed to repeat the sequence.



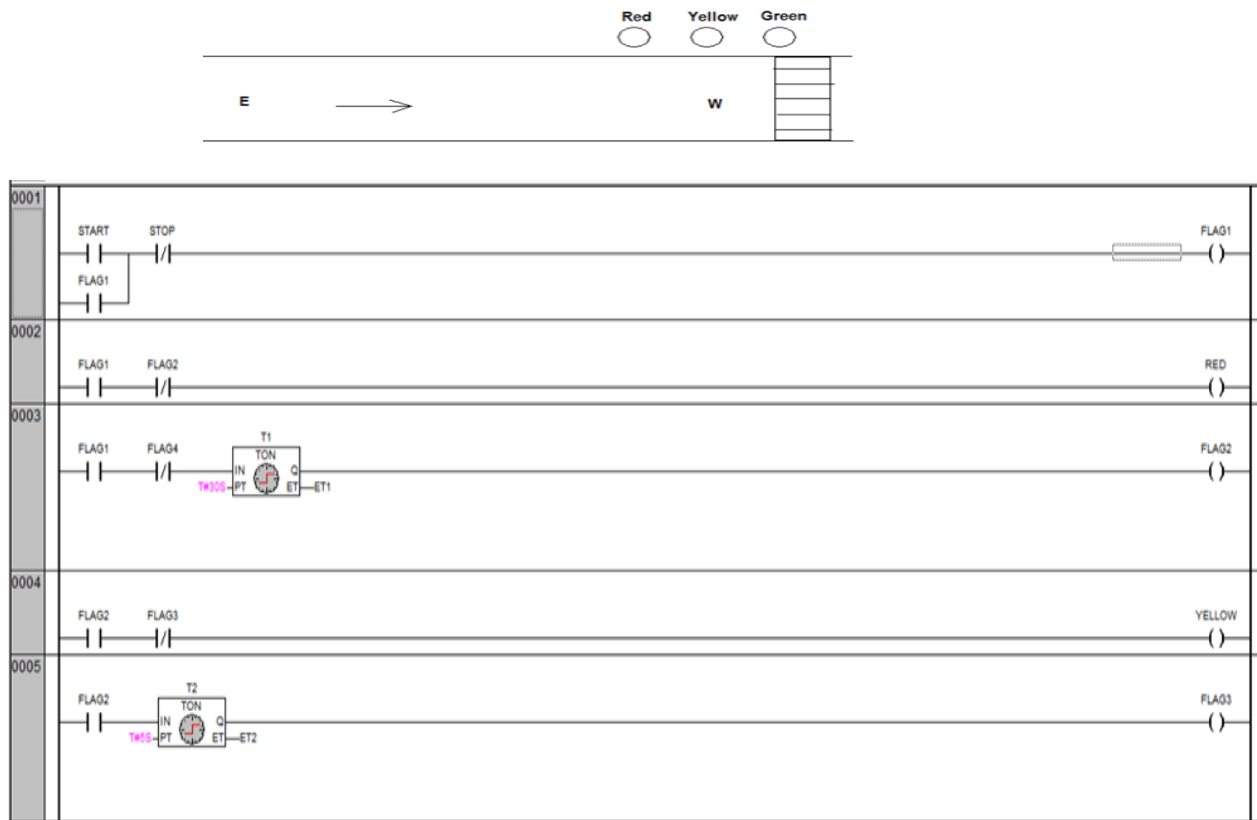
19. When the lights are turned off in a building, an exit door light is to remain on for an additional 2 min, and the parking lot lights are to remain on for an additional 3 min after the door light goes out. Write a program to implement this process.

20. Name the three forms of PLC counter instructions, and explain the basic operation of each.

21. When is the output of a PLC counter energized?

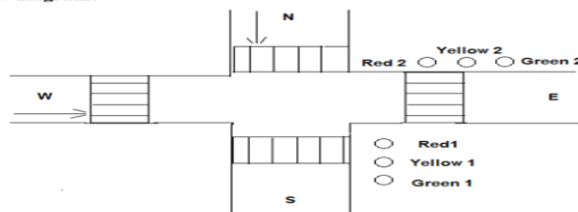
22. Write a program that will increment a counter's accumulated value 1 count every 60 s. A second counter's accumulated value will increment 1 count every time the first counter's accumulated value reaches 60. The first counter will reset when its accumulated value reaches 60, and the second counter will reset when its accumulated value reaches 12.

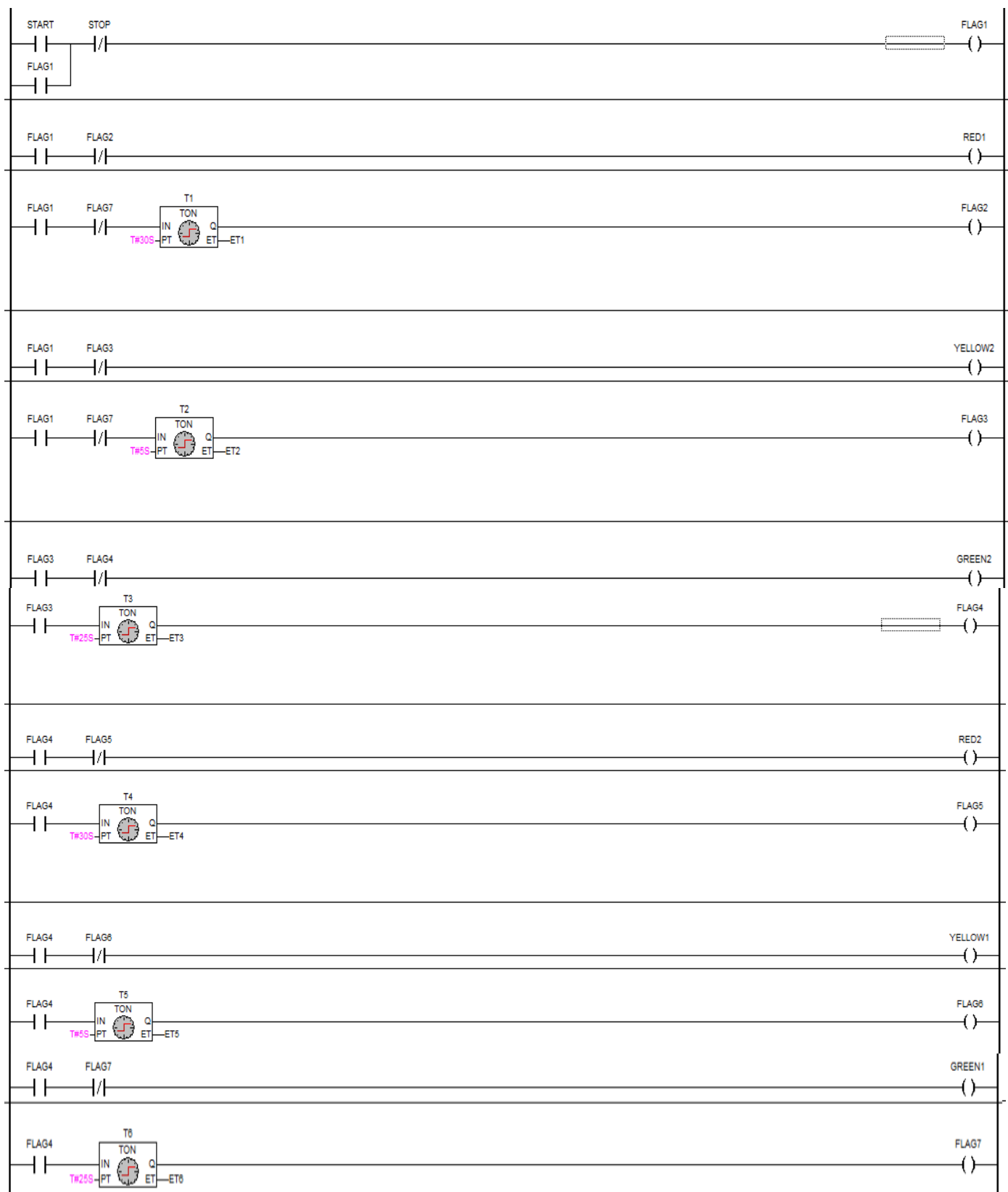
23. Develop the ladder diagram for a one way traffic light control system. When start button is pressed, Red light will be ON for 30s after which yellow light is ON for 5s then green light is ON for 25s and this process continues. There is a start and stop button to control the system.



24. When Red 1 is ON for 30s, E-W traffic flows. At this instant, Yellow 2 becomes ON for 5s after which Green 2 becomes ON for 25s. After 30s, when Red 2 is ON for 30s, N-S traffic flows. At this instant, Yellow 1 becomes ON for 5s after which Green 1 becomes ON for 25s and this process continues. There is a start button and a stop button. Draw the ladder logic to implement the above process.

Flow diagram:





25. Write a PLC program to operate a light according to the following sequence.

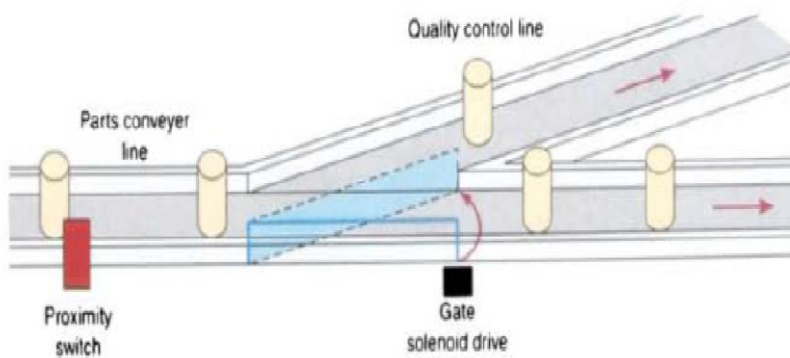
- A momentary push button is pressed to start the sequence
- The light is switched ON and remains ON for 2 s.
- The light is then switched off and remains off for 2 s.
- A counter is incremented by 1 after this sequence.
- The sequence then repeats for a total of 4 counts.
- After the fourth count, the sequence will stop and the counter will be reset to zero.



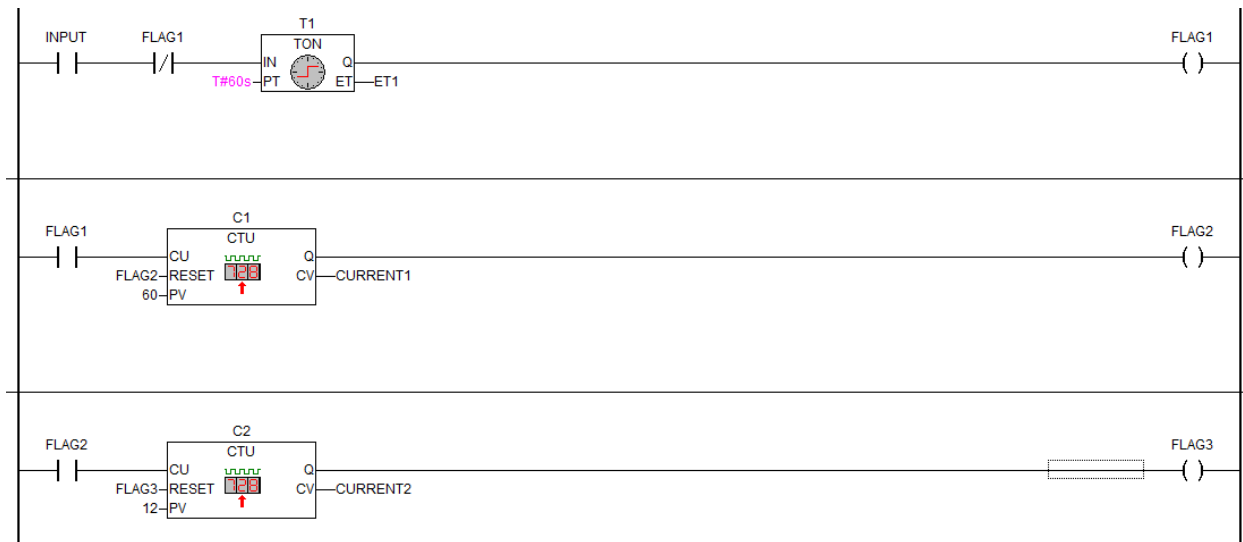
26. Three conveyors feed a main conveyor. The count from each feeder conveyor is fed into an input register in the PLC. Construct a PLC program to obtain the total count of parts on the main conveyor. Use a timer to update the total every 15 seconds. Design, construct, and test PLC circuits for this process
27. In certain process control application o/p is ON if the count is less than 34 or more than 41. Implement the same using PLC ladder diagram
28. An indicating light is to go ON when a count reaches 23. The light is then go off when a count of 31 is reached. Design, construct, and test PLC circuits for this process
29. In certain process control application when the count reaches 25, a paint spray is to run for 40 seconds. Design, construct, and test PLC circuits for this process.
30. Write a program to implement the process illustrated in Figure below. An up-counter must be programmed as part of a batch-counting operation to sort parts automatically for quality

control. The counter is installed to divert 1 part out of every 1000 for quality control or inspection purposes. The circuit operates as follows:

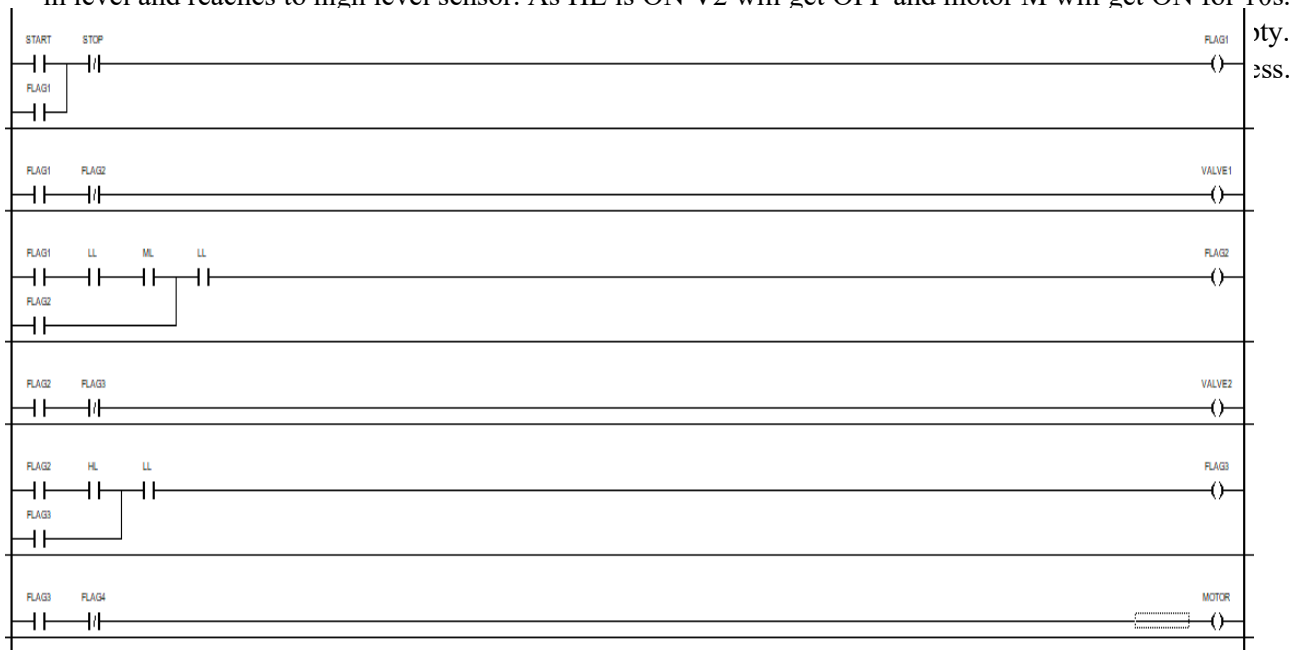
- A start/stop pushbutton station is used to turn the conveyor motor on and off.
- A proximity sensor counts the parts as they pass by on the conveyor.
- When a count of 1000 is reached, the counter's output activates *the* gate solenoid, diverting the part to the inspection line.
- The gate solenoid is energized for 2 s, which allows enough time for the part to continue to the quality control line.
- The gate returns to its normal position when the 2-s time period ends.
- The counter resets to 0 and continues to accumulate counts.
- A reset pushbutton is provided to reset the counter manually.



Write a program that will increment a counter's accumulated value 1 count every 60 s. A second counter's accumulated value will increment 1 count every time the first counter's accumulated value reaches 60. The first counter will reset when its accumulated value reaches 60, and the second counter will reset when its accumulated value reaches 12.



When START is pressed the valve V1 will get ON resulting liquid1 inlet to mixing tank. As liquid1 comes in low level sensor will get ON and further middle level sensor will be ON. As ML is ON, V1 should get OFF and V2 should get ON. As V2 is ON liquid 2 inlet to mixing tank resulting increase in level and reaches to high level sensor. As HL is ON V2 will get OFF and motor M will get ON for 10s.



Module-3 - SCADA and DCS

1. List the elements of SCADA
2. Why is an uninterrupted power supply needed at an RTU?
3. Explain briefly about various functional elements of SCADA system
4. Describe how MTU handles analog control instructions to open and close a valve in a SCADA system.
5. List the signals that come into the RTU
6. What are the main functions of a master terminal unit of a SCADA?
7. Explain briefly about the analog input and output module of a SCADA system
8. Describe how MTU handles discrete control instructions to open and close a valve in a SCADA system.
9. List the signals that leave the RTU
10. Many protocols divide the binary message in to three fields. What are they?
11. Explain briefly about the digital input and output module of a SCADA system
12. Discuss briefly about the functional elements of a Remote Terminal Unit (RTU)
13. Define control complexity ratio of a control system structure.
14. Compare different control system architectures on its features
15. Describe briefly about the architecture of a DCS
16. Explain the design considerations and requirements of operator interface in detail.
17. Explain the architectural parameters of LCU and compare its different configurations.
18. Describe briefly about the hybrid system architecture
19. Recall the use of function blocks in programming LCU
20. Sketch the architecture of geographically centralized control configuration.
21. Compare various configuration of LCUs on its architectural parameters.
22. List few signal status functions used in a function block
23. Explain the function block programming approach with an example
24. Describe briefly about the architecture of a DCS